C.6 Hypothesis Testing

So far, we have reviewed how to evaluate point estimators, and we have seen—in the case of a population mean—how to construct and interpret confidence intervals. But sometimes the question we are interested in has a definite yes or no answer. Here are some examples: (1) Does a job training program effectively increase average worker productivity? (see Example C.2); (2) Are blacks discriminated against in hiring? (see Example C.3); (3) Do stiffer state drunk driving laws reduce the number of drunk driving arrests? Devising methods for answering such questions, using a sample of data, is known as hypothesis testing.

Fundamentals of Hypothesis Testing

To illustrate the issues involved with hypothesis testing, consider an election example. Suppose there are two candidates in an election, Candidates A and B. Candidate A is reported to have received 42% of the popular vote, while Candidate B received 58%. These are supposed to represent the true percentages in the voting population, and we treat them as such.

Candidate A is convinced that more people must have voted for him, so he would like to investigate whether the election was rigged. Knowing something about statistics, Candidate A hires a consulting agency to randomly sample 100 voters to record whether or not each person voted for him. Suppose that, for the sample collected, 53 people voted for Candidate A. This sample estimate of 53% clearly exceeds the reported population value of 42%. Should Candidate A conclude that the election was indeed a fraud?

While it appears that the votes for Candidate A were undercounted, we cannot be certain. Even if only 42% of the population voted for Candidate A, it is possible that, in a sample of 100, we observe 53 people who did vote for Candidate A. The question is: How *strong* is the sample evidence against the officially reported percentage of 42%?

One way to proceed is to set up a **hypothesis test**. Let θ denote the true proportion of the population voting for Candidate A. The hypothesis that the reported results are accurate can be stated as

$$H_0: \theta = .42. \tag{C.28}$$

This is an example of a **null hypothesis**. We always denote the null hypothesis by H_0. In hypothesis testing, the null hypothesis plays a role similar to that of a defendant on trial in many judicial systems: just as a defendant is presumed to be innocent until proven guilty, the null hypothesis is presumed to be true until the data strongly suggest otherwise. In the current example, Candidate A must present fairly strong evidence against (C.28) in order to win a recount.

The **alternative hypothesis** in the election example is that the true proportion voting for Candidate A in the election is greater than .42:

$$H_1: \theta > .42. \tag{C.29}$$

In order to conclude that H_0 is false and that H_1 is true, we must have evidence "beyond reasonable doubt" against H_0. How many votes out of 100 would be needed before we feel the evidence is strongly against H_0? Most would agree that observing 43 votes out of a sample of 100 is not enough to overturn the original election results; such an outcome is well within the expected sampling variation. On the other hand, we do not need to observe 100 votes for Candidate A to cast doubt on H_0. Whether 53 out of 100 is enough to reject H_0 is much less clear. The answer depends on how we quantify "beyond reasonable doubt."

Before we turn to the issue of quantifying uncertainty in hypothesis testing, we should head off some possible confusion. You may have noticed that the hypotheses in equations (C.28) and (C.29) do not exhaust all possibilities: it could be that θ is less than .42. For the application at hand, we are not particularly interested in that possibility; it has nothing to do with overturning the results of the election. Therefore, we can just state at the outset that we are ignoring alternatives θ with $\theta < .42$. Nevertheless, some authors prefer to state null and alternative hypotheses so that they are exhaustive, in which case our null hypothesis should be H_0: $\theta \leq .42$. Stated in this way, the null hypothesis is a *composite* null hypothesis because it allows for more than one value under H_0. [By contrast, equation (C.28) is an example of a *simple* null hypothesis.] For these kinds of examples, it does not matter whether we state the null as in (C.28) or as a composite null: the most difficult value to reject if $\theta \leq .42$ is $\theta = .42$. (That is, if we reject the value $\theta = .42$, against $\theta > .42$, then logically we must reject any value less than .42.) Therefore, our testing procedure based on (C.28) leads to the same test as if H_0: $\theta \leq .42$. In this text, we always state a null hypothesis as a simple null hypothesis.

In hypothesis testing, we can make two kinds of mistakes. First, we can reject the null hypothesis when it is in fact true. This is called a **Type I error**. In the election example, a Type I error occurs if we reject H_0 when the true proportion of people voting for Candidate A is in fact .42. The second kind of error is failing to reject H_0 when it is actually false. This is called a **Type II error**. In the election example, a Type II error occurs if $\theta > .42$ but we fail to reject H_0.

After we have made the decision of whether or not to reject the null hypothesis, we have either decided correctly or we have committed an error. We will never know with certainty whether an error was committed. However, we can compute the *probability* of making either a Type I or a Type II error. Hypothesis testing rules are constructed to make the probability of committing a Type I error fairly small. Generally, we define the **significance level** (or simply the *level*) of a test as the probability of a Type I error; it is typically denoted by α. Symbolically, we have

$$\alpha = \mathrm{P}(\text{Reject } H_0 | H_0). \qquad \text{C.30}$$

The right-hand side is read as: "The probability of rejecting H_0 given that H_0 is true."

Classical hypothesis testing requires that we initially specify a significance level for a test. When we specify a value for α, we are essentially quantifying our tolerance for a Type I error. Common values for α are .10, .05, and .01. If $\alpha = .05$, then the researcher is willing to falsely reject H_0 5% of the time, in order to detect deviations from H_0.

Once we have chosen the significance level, we would then like to minimize the probability of a Type II error. Alternatively, we would like to maximize the **power of a test** against all relevant alternatives. The power of a test is just one minus the probability of a Type II error. Mathematically,

$$\pi(\theta) = \mathrm{P}(\text{Reject } H_0 | \theta) = 1 - \mathrm{P}(\text{Type II} | \theta),$$

where θ denotes the actual value of the parameter. Naturally, we would like the power to equal unity whenever the null hypothesis is false. But this is impossible to achieve while keeping the significance level small. Instead, we choose our tests to maximize the power for a given significance level.

Testing Hypotheses about the Mean in a Normal Population

In order to test a null hypothesis against an alternative, we need to choose a test statistic (or statistic, for short) and a critical value. The choices for the statistic and critical value are based on convenience and on the desire to maximize power given a significance level for the test. In this subsection, we review how to test hypotheses for the mean of a normal population.

A **test statistic**, denoted T, is some function of the random sample. When we compute the statistic for a particular outcome, we obtain an outcome of the test statistic, which we will denote t.

Given a test statistic, we can define a rejection rule that determines when H_0 is rejected in favor of H_1. In this text, all rejection rules are based on comparing the value of a test statistic, t, to a **critical value**, c. The values of t that result in rejection of the null hypothesis are collectively known as the **rejection region**. To determine the critical value, we must first decide on a significance level of the test. Then, given α, the critical value associated with α is determined by the distribution of T, *assuming* that H_0 is true. We will write this critical value as c, suppressing the fact that it depends on α.

Testing hypotheses about the mean μ from a Normal(μ,σ^2) population is straightforward. The null hypothesis is stated as

$$H_0\colon \mu = \mu_0, \tag{C.31}$$

where μ_0 is a value that we specify. In the majority of applications, $\mu_0 = 0$, but the general case is no more difficult.

The rejection rule we choose depends on the nature of the alternative hypothesis. The three alternatives of interest are

$$H_1\colon \mu > \mu_0, \tag{C.32}$$

$$H_1\colon \mu < \mu_0, \tag{C.33}$$

and

$$H_1\colon \mu \neq \mu_0. \tag{C.34}$$

Equation (C.32) gives a **one-sided alternative**, as does (C.33). When the alternative hypothesis is (C.32), the null is effectively $H_0\colon \mu \leq \mu_0$, since we reject H_0 only when $\mu > \mu_0$. This is appropriate when we are interested in the value of μ only when μ is at least as large as μ_0. Equation (C.34) is a **two-sided alternative**. This is appropriate when we are interested in any departure from the null hypothesis.

Consider first the alternative in (C.32). Intuitively, we should reject H_0 in favor of H_1 when the value of the sample average, $\bar{y}$, is "sufficiently" greater than μ_0. But how should we determine when $\bar{y}$ is large enough for H_0 to be rejected at the chosen significance level? This requires knowing the probability of rejecting the null hypothesis when it is true. Rather than working directly with $\bar{y}$, we use its standardized version, where σ is replaced with the sample standard deviation, s:

$$t = \sqrt{n}(\bar{y} - \mu_0)/s = (\bar{y} - \mu_0)/\mathrm{se}(\bar{y}), \tag{C.35}$$

where $\text{se}(\bar{y}) = s/\sqrt{n}$ is the standard error of $\bar{y}$. Given the sample of data, it is easy to obtain t. We work with t because, under the null hypothesis, the random variable

$$T = \sqrt{n}(\bar{Y} - \mu_0)/S$$

has a t_{n-1} distribution. Now, suppose we have settled on a 5% significance level. Then, the critical value c is chosen so that $P(T > c|H_0) = .05$; that is, the probability of a Type I error is 5%. Once we have found c, the rejection rule is

$$t > c, \tag{C.36}$$

where c is the $100(1 - \alpha)$ percentile in a t_{n-1} distribution; as a percent, the significance level is $100\cdot\alpha\%$. This is an example of a **one-tailed test** because the rejection region is in one tail of the t distribution. For a 5% significance level, c is the 95$^{\text{th}}$ percentile in the t_{n-1} distribution; this is illustrated in Figure C.5. A different significance level leads to a different critical value.

The statistic in equation (C.35) is often called the ***t* statistic** for testing H_0: $\mu = \mu_0$. The t statistic measures the distance from $\bar{y}$ to μ_0 *relative* to the standard error of $\bar{y}$, $\text{se}(\bar{y})$.

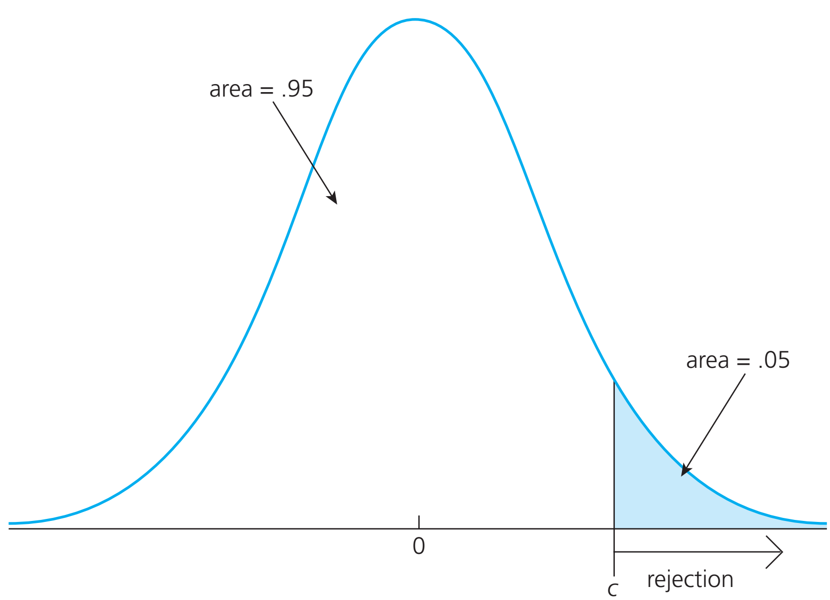
Example C.4

[Effect of Enterprise Zones on Business Investments]

In the population of cities granted enterprise zones in a particular state [see Papke (1994) for Indiana], let Y denote the percentage change in investment from the year before to the year after a city became

FIGURE C.5

Rejection region for a 5% significance level test against the one-sided alternative $\mu > \mu_0$.



an enterprise zone. Assume that Y has a Normal(μ,σ^2) distribution. The null hypothesis that enterprise zones have no effect on business investment is H_0: $\mu = 0$; the alternative that they have a positive effect is H_1: $\mu > 0$. (We assume that they do not have a negative effect.) Suppose that we wish to test H_0 at the 5% level. The test statistic in this case is

$$t = \frac{\bar{y}}{s/\sqrt{n}} = \frac{\bar{y}}{\text{se}(\bar{y})}. \tag{C.37}$$

Suppose that we have a sample of 36 cities that are granted enterprise zones. Then, the critical value is $c = 1.69$ (see Table G.2), and we reject H_0 in favor of H_1 if $t > 1.69$. Suppose that the sample yields $\bar{y} = 8.2$ and $s = 23.9$. Then, $t \approx 2.06$, and H_0 is therefore rejected at the 5% level. Thus, we conclude that, at the 5% significance level, enterprise zones have an effect on average investment. The 1% critical value is 2.44, so H_0 is not rejected at the 1% level. The same caveat holds here as in Example C.2: we have not controlled for other factors that might affect investment in cities over time, so we cannot claim that the effect is causal.

The rejection rule is similar for the one-sided alternative (C.33). A test with a significance level of $100\cdot\alpha\%$ rejects H_0 against (C.33) whenever

$$t < -c; \tag{C.38}$$

in other words, we are looking for negative values of the t statistic—which implies $\bar{y} < \mu_0$—that are sufficiently far from zero to reject H_0.

For two-sided alternatives, we must be careful to choose the critical value so that the significance level of the test is still α. If H_1 is given by H_1: $\mu \neq \mu_0$, then we reject H_0 if $\bar{y}$ is far from μ_0 in *absolute value*: a $\bar{y}$ much larger or much smaller than μ_0 provides evidence against H_0 in favor of H_1. A $100\cdot\alpha\%$ level test is obtained from the rejection rule

$$|t| > c, \tag{C.39}$$

where $|t|$ is the absolute value of the t statistic in (C.35). This gives a **two-tailed test**. We must now be careful in choosing the critical value: c is the $100(1 - \alpha/2)$ percentile in the t_{n-1} distribution. For example, if $\alpha = .05$, then the critical value is the 97.5th percentile in the t_{n-1} distribution. This ensures that H_0 is rejected only 5% of the time when it is true (see Figure C.6). For example, if $n = 22$, then the critical value is $c = 2.08$, the 97.5th percentile in a t_{21} distribution (see Table G.2). The absolute value of the t statistic must exceed 2.08 in order to reject H_0 against H_1 at the 5% level.

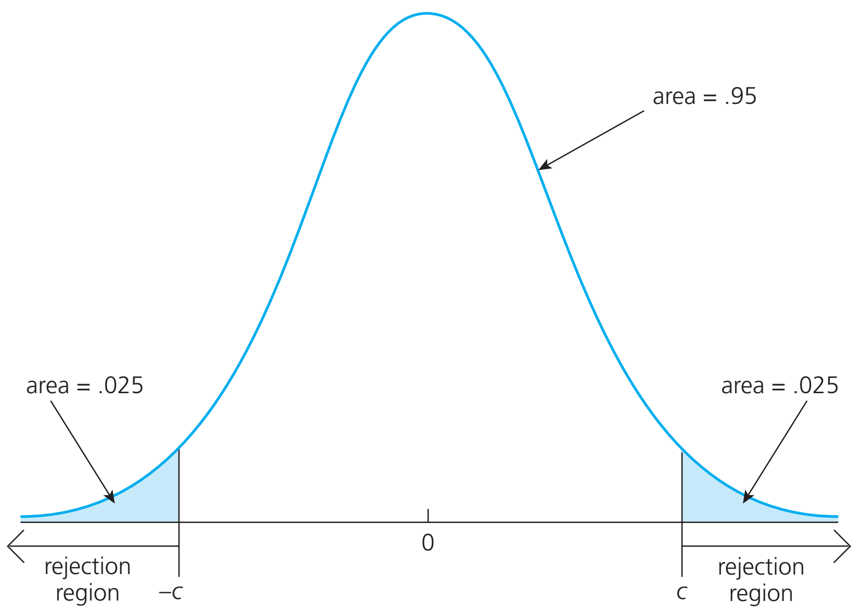
It is important to know the proper language of hypothesis testing. Sometimes, the appropriate phrase "we fail to reject H_0 in favor of H_1 at the 5% significance level" is replaced with "we accept H_0 at the 5% significance level." The latter wording is incorrect. With the same set of data, there are usually many hypotheses that cannot be rejected. In the earlier election example, it would be logically inconsistent to say that H_0: $\theta = .42$ and H_0: $\theta = .43$ are both "accepted," since only one of these can be true. But it is entirely possible that neither of these hypotheses is rejected. For this reason, we always say "fail to reject H_0" rather than "accept H_0."

Asymptotic Tests for Nonnormal Populations

If the sample size is large enough to invoke the central limit theorem (see Section C.3), the mechanics of hypothesis testing for population means are the *same* whether or not the

FIGURE C.6

Rejection region for a 5% significance level test against the two-sided alternative H_1: $\mu \neq \mu_0$.



population distribution is normal. The theoretical justification comes from the fact that, under the null hypothesis,

$$T = \sqrt{n}(\bar{Y} - \mu_0)/S \overset{a}{\sim} \text{Normal}(0,1).$$

Therefore, with large n, we can compare the t statistic in (C.35) with the critical values from a standard normal distribution. Because the t_{n-1} distribution converges to the standard normal distribution as n gets large, the t and standard normal critical values will be very close for extremely large n. Because asymptotic theory is based on n increasing without bound, it cannot tell us whether the standard normal or t critical values are better. For moderate values of n, say, between 30 and 60, it is traditional to use the t distribution because we know this is correct for normal populations. For $n > 120$, the choice between the t and standard normal distributions is largely irrelevant because the critical values are practically the same.

Because the critical values chosen using either the standard normal or t distribution are only approximately valid for nonnormal populations, our chosen significance levels are also only approximate; thus, for nonnormal populations, our significance levels are really *asymptotic* significance levels. Thus, if we choose a 5% significance level, but our population is nonnormal, then the actual significance level will be larger or smaller than 5% (and we cannot know which is the case). When the sample size is large, the actual significance level will be very close to 5%. Practically speaking, the distinction is not important, so we will now drop the qualifier "asymptotic."

Example C.5

[Race Discrimination in Hiring]

In the Urban Institute study of discrimination in hiring (see Example C.3), we are primarily interested in testing H_0: $\mu = 0$ against H_1: $\mu < 0$, where $\mu = \theta_B - \theta_W$ is the difference in probabilities that blacks and whites receive job offers. Recall that μ is the population mean of the variable $Y = B - W$, where B and W are binary indicators. Using the $n = 241$ paired comparisons, we obtained $\bar{y} = -.133$ and $\text{se}(\bar{y}) = .482/\sqrt{241} \approx .031$. The t statistic for testing H_0: $\mu = 0$ is $t = -.133/.031 \approx -4.29$. You will remember from Appendix B that the standard normal distribution is, for practical purposes, indistinguishable from the t distribution with 240 degrees of freedom. The value -4.29 is so far out in the left tail of the distribution that we reject H_0 at any reasonable significance level. In fact, the .005 (one-half of a percent) critical value (for the one-sided test) is about -2.58. A t value of -4.29 is *very* strong evidence against H_0 in favor of H_1. Hence, we conclude that there is discrimination in hiring.

Computing and Using *p*-Values

The traditional requirement of choosing a significance level ahead of time means that different researchers, using the same data and same procedure to test the same hypothesis, could wind up with different conclusions. Reporting the significance level at which we are carrying out the test solves this problem to some degree, but it does not completely remove the problem.

To provide more information, we can ask the following question: What is the *largest* significance level at which we could carry out the test and still fail to reject the null hypothesis? This value is known as the ***p*-value** of a test (sometimes called the *prob-value*). Compared with choosing a significance level ahead of time and obtaining a critical value, computing a p-value is somewhat more difficult. But with the advent of quick and inexpensive computing, p-values are now fairly easy to obtain.

As an illustration, consider the problem of testing H_0: $\mu = 0$ in a Normal(μ,σ^2) population. Our test statistic in this case is $T = \sqrt{n}\cdot\bar{Y}/S$, and we assume that n is large enough to treat T as having a standard normal distribution under H_0. Suppose that the observed value of T for our sample is $t = 1.52$. (Note how we have skipped the step of choosing a significance level.) Now that we have seen the value t, we can find the largest significance level at which we would fail to reject H_0. This is the significance level associated with using t as our critical value. Because our test statistic T has a standard normal distribution under H_0, we have

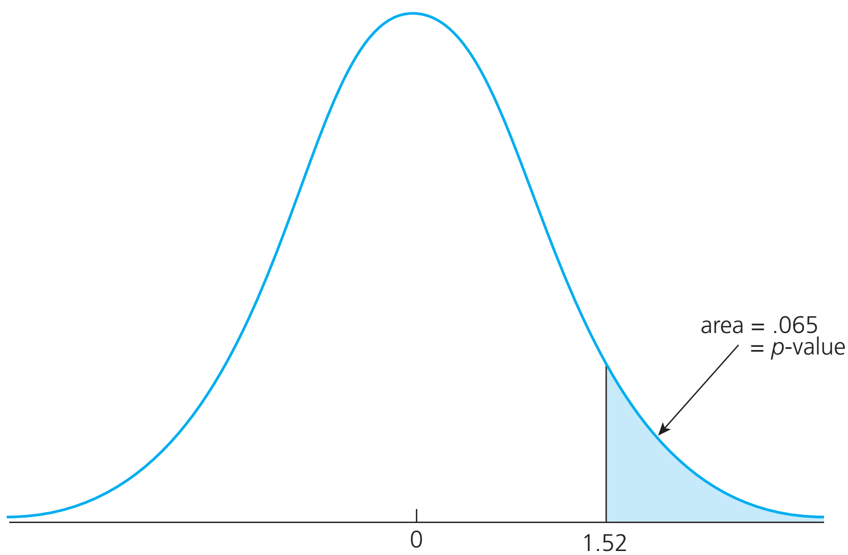
$$p\text{-value} = P(T > 1.52|H_0) = 1 - \Phi(1.52) = .065, \tag{C.40}$$

where $\Phi(\cdot)$ denotes the standard normal cdf. In other words, the p-value in this example is simply the area to the right of 1.52, the observed value of the test statistic, in a standard normal distribution. See Figure C.7 for illustration.

Because p-value = .065, the largest significance level at which we can carry out this test and fail to reject is 6.5%. If we carry out the test at a level below 6.5% (such as at 5%), we fail to reject H_0. If we carry out the test at a level larger than 6.5% (such as 10%), we reject H_0. With the p-value at hand, we can carry out the test at any level.

FIGURE C.7

The p-value when $t = 1.52$ for the one-sided alternative $\mu > \mu_0$.



The p-value in this example has another useful interpretation: it is the probability that we observe a value of T as large as 1.52 when the null hypothesis is true. If the null hypothesis is actually true, we would observe a value of T as large as 1.52 due to chance only 6.5% of the time. Whether this is small enough to reject H_0 depends on our tolerance for a Type I error. The p-value has a similar interpretation in all other cases, as we will see.

Generally, small p-values are evidence *against* H_0, since they indicate that the outcome of the data occurs with small probability if H_0 is true. In the previous example, if t had been a larger value, say, $t = 2.85$, then the p-value would be $1 - \Phi(2.85) \approx .002$. This means that, if the null hypothesis were true, we would observe a value of T as large as 2.85 with probability .002. How do we interpret this? Either we obtained a very unusual sample or the null hypothesis is false. Unless we have a *very* small tolerance for Type I error, we would reject the null hypothesis. On the other hand, a large p-value is weak evidence against H_0. If we had gotten $t = .47$ in the previous example, then p-value $= 1 - \Phi(.47) = .32$. Observing a value of T larger than .47 happens with probability .32, even when H_0 is true; this is large enough so that there is insufficient doubt about H_0, unless we have a very high tolerance for Type I error.

For hypothesis testing about a population mean using the t distribution, we need detailed tables in order to compute p-values. Table G.2 only allows us to put bounds on p-values. Fortunately, many statistics and econometrics packages now compute p-values routinely, and they also provide calculation of cdfs for the t and other distributions used for computing p-values.

Example C.6

[Effect of Job Training Grants on Worker Productivity]

Consider again the Holzer et al. (1993) data in Example C.2. From a policy perspective, there are two questions of interest. First, what is our best estimate of the mean change in scrap rates, μ? We have already obtained this for the sample of 20 firms listed in Table C.3: the sample average of the change in scrap rates is -1.15. Relative to the initial average scrap rate in 1987, this represents a fall in the scrap rate of about 26.3% ($-1.15/4.38 \approx -.263$), which is a nontrivial effect.

We would also like to know whether the sample provides strong evidence for an effect in the population of manufacturing firms that could have received grants. The null hypothesis is H_0: $\mu = 0$, and we test this against H_1: $\mu < 0$, where μ is the average change in scrap rates. Under the null, the job training grants have no effect on average scrap rates. The alternative states that there is an effect. We do not care about the alternative $\mu > 0$, so the null hypothesis is effectively H_0: $\mu \geq 0$.

Since $\bar{y} = -1.15$ and $\text{se}(\bar{y}) = .54$, $t = -1.15/.54 = -2.13$. This is below the 5% critical value of -1.73 (from a t_{19} distribution) but above the 1% critical value, -2.54. The p-value in this case is computed as

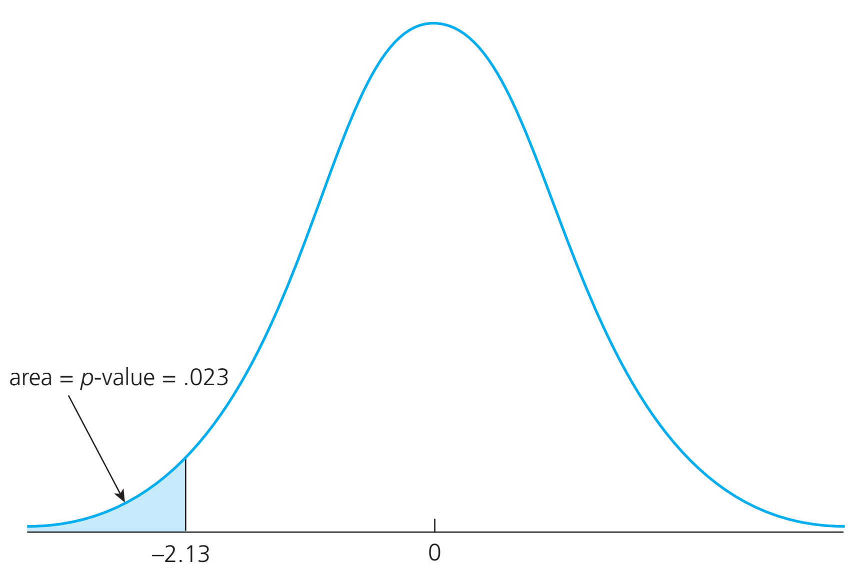
$$p\text{-value} = \text{P}(T_{19} < -2.13), \tag{C.41}$$

where T_{19} represents a t distributed random variable with 19 degrees of freedom. The inequality is reversed from (C.40) because the alternative has the form in (C.33). The probability in (C.41) is the area to the left of -2.13 in a t_{19} distribution (see Figure C.8).

Using Table G.2, the most we can say is that the p-value is between .025 and .01, but it is closer to .025 (since the 97.5th percentile is about 2.09). Using a statistical package, such as Stata, we can

FIGURE C.8

The p-value when $t = -2.13$ with 19 degrees of freedom for the one-sided alternative $\mu < 0$.



compute the exact p-value. It turns out to be about .023, which is reasonable evidence against H_0. This is certainly enough evidence to reject the null hypothesis that the training grants had no effect at the 2.5% significance level (and therefore at the 5% level).

Computing a p-value for a two-sided test is similar, but we must account for the two-sided nature of the rejection rule. For t testing about population means, the p-value is computed as

$$P(|T_{n-1}| > |t|) = 2P(T_{n-1} > |t|), \tag{C.42}$$

where t is the value of the test statistic and T_{n-1} is a t random variable. (For large n, replace T_{n-1} with a standard normal random variable.) Thus, compute the absolute value of the t statistic, find the area to the right of this value in a t_{n-1} distribution, and multiply the area by two.

For nonnormal populations, the exact p-value can be difficult to obtain. Nevertheless, we can find *asymptotic* p-values by using the same calculations. These p-values are valid for large sample sizes. For n larger than, say, 120, we might as well use the standard normal distribution. Table G.1 is detailed enough to get accurate p-values, but we can also use a statistics or econometrics program.

Example C.7

[Race Discrimination in Hiring]

Using the matched pair data from the Urban Institute ($n = 241$), we obtained $t = -4.29$. If Z is a standard normal random variable, $P(Z < -4.29)$ is, for practical purposes, zero. In other words, the (asymptotic) p-value for this example is essentially zero. This is very strong evidence against H_0.

Summary of How to Use *p*-Values:

(i) Choose a test statistic T and decide on the nature of the alternative. This determines whether the rejection rule is $t > c$, $t < -c$, or $|t| > c$.

(ii) Use the observed value of the t statistic as the critical value and compute the corresponding significance level of the test. This is the p-value. If the rejection rule is of the form $t > c$, then p-value $= P(T > t)$. If the rejection rule is $t < -c$, then p-value $= P(T < t)$; if the rejection rule is $|t| > c$, then p-value $= P(|T| > |t|)$.

(iii) If a significance level α has been chosen, then we reject H_0 at the $100\cdot\alpha\%$ level if p-value $< \alpha$. If p-value $\geq \alpha$, then we fail to reject H_0 at the $100\cdot\alpha\%$ level. Therefore, it is a small p-value that leads to rejection.

The Relationship between Confidence Intervals and Hypothesis Testing

Because contructing confidence intervals and hypothesis tests both involve probability statements, it is natural to think that they are somehow linked. It turns out that they are. After a confidence interval has been constructed, we can carry out a variety of hypothesis tests.

The confidence intervals we have discussed are all two-sided by nature. (In this text, we will have no need to construct one-sided confidence intervals.) Thus, confidence intervals can be used to test against *two-sided* alternatives. In the case of a population mean, the null is given by (C.31), and the alternative is (C.34). Suppose we have constructed a 95% confidence interval for μ. Then, if the hypothesized value of μ under H_0, μ_0, is not in the confidence interval, then $H_0: \mu = \mu_0$ is rejected against $H_1: \mu \neq \mu_0$ at the 5% level. If μ_0 lies in this interval, then we fail to reject H_0 at the 5% level. Notice how any value for μ_0 can be tested once a confidence interval is constructed, and since a confidence interval contains more than one value, there are many null hypotheses that will not be rejected.

Example C.8

[Training Grants and Worker Productivity]

In the Holzer et al. example, we constructed a 95% confidence interval for the mean change in scrap rate μ as $[-2.28, -.02]$. Since zero is excluded from this interval, we reject $H_0: \mu = 0$ against $H_1: \mu \neq 0$ at the 5% level. This 95% confidence interval also means that we fail to reject $H_0: \mu = -2$ at the 5% level. In fact, there is a continuum of null hypotheses that are not rejected given this confidence interval.

Practical versus Statistical Significance

In the examples covered so far, we have produced three kinds of evidence concerning population parameters: point estimates, confidence intervals, and hypothesis tests. These tools for learning about population parameters are equally important. There is an understandable tendency for students to focus on confidence intervals and hypothesis tests because these are things to which we can attach confidence or significance levels. But in any study, we must also interpret the *magnitudes* of point estimates.

The sign and magnitude of $\bar{y}$ determine its **practical significance** and allow us to discuss the direction of an intervention or policy effect, and whether the estimated effect is "large" or "small." On the other hand, **statistical significance** of $\bar{y}$ depends on the magnitude of its t statistic. For testing $H_0: \mu = 0$, the t statistic is simply $t = \bar{y}/\text{se}(\bar{y})$. In other words, statistical significance depends on the ratio of $\bar{y}$ to its standard error. Consequently, a t statistic can be large because $\bar{y}$ is large or $\text{se}(\bar{y})$ is small. In applications, it is important to discuss both practical and statistical significance, being aware that an estimate can be statistically significant without being especially large in a practical sense. Whether an estimate is practically important depends on the context as well as on one's judgment, so there are no set rules for determining practical significance.

Example C.9

[Effect of Freeway Width on Commute Time]

Let Y denote the change in commute time, measured in minutes, for commuters in a metropolitan area from before a freeway was widened to after the freeway was widened. Assume that $Y \sim \text{Normal}(\mu,\sigma^2)$. The null hypothesis that the widening did not reduce average commute time is $H_0: \mu = 0$; the alternative that it reduced average commute time is $H_1: \mu < 0$. Suppose a random sample of commuters of size $n = 900$ is obtained to determine the effectiveness of the freeway project. The average change in commute time is computed to be $\bar{y} = -3.6$, and the sample standard deviation is $s = 32.7$; thus, $\text{se}(\bar{y}) = 32.7/\sqrt{900} = 1.09$. The t statistic is $t = -3.6/1.09 \approx -3.30$, which is very statistically significant; the p-value is about .0005. Thus, we conclude that the freeway widening had a statistically significant effect on average commute time.

If the outcome of the hypothesis test is all that were reported from the study, it would be misleading. Reporting only statistical significance masks the fact that the estimated reduction in average commute time, 3.6 minutes, is pretty meager. To be up front, we should report the point estimate of -3.6, along with the significance test.

Finding point estimates that are statistically significant without being practically significant can occur when we are working with large samples. To discuss why this happens, it is useful to have the following definition.

Test Consistency. A **consistent test** rejects H_0 with probability approaching one as the sample size grows whenever H_1 is true.

Another way to say that a test is consistent is that, as the sample size tends to infinity, the power of the test gets closer and closer to unity whenever H_1 is true. All of the tests we cover in this text have this property. In the case of testing hypotheses about a population mean, test consistency follows because the variance of $\bar{Y}$ converges to zero as the sample size gets large. The t statistic for testing H_0: $\mu = 0$ is $T = \bar{Y}/(S/\sqrt{n})$. Since $\text{plim}(\bar{Y}) = \mu$ and $\text{plim}(S) = \sigma$, it follows that if, say, $\mu > 0$, then T gets larger and larger (with high probability) as $n \to \infty$. In other words, no matter how close μ is to zero, we can be almost certain to reject H_0: $\mu = 0$ given a large enough sample size. This says nothing about whether μ is large in a practical sense.

C.7 Remarks on Notation

In our review of probability and statistics here and in Appendix B, we have been careful to use standard conventions to denote random variables, estimators, and test statistics. For example, we have used W to indicate an estimator (random variable) and w to denote a particular estimate (outcome of the random variable W). Distinguishing between an estimator and an estimate is important for understanding various concepts in estimation and hypothesis testing. However, making this distinction quickly becomes a burden in econometric analysis because the models are more complicated: many random variables and parameters will be involved, and being true to the usual conventions from probability and statistics requires many extra symbols.

In the main text, we use a simpler convention that is widely used in econometrics. If θ is a population parameter, the notation $\hat{\theta}$ ("theta hat") will be used to denote both an estimator and an estimate of θ. This notation is useful in that it provides a simple way of attaching an estimator to the population parameter it is supposed to be estimating. Thus, if the population parameter is β, then $\hat{\beta}$ denotes an estimator or estimate of β; if the parameter is σ^2, $\hat{\sigma}^2$ is an estimator or estimate of σ^2; and so on. Sometimes, we will discuss two estimators of the same parameter, in which case we will need a different notation, such as $\tilde{\theta}$ ("theta tilde").

Although dropping the conventions from probability and statistics to indicate estimators, random variables, and test statistics puts additional responsibility on you, it is not a big deal once the difference between an estimator and an estimate is understood. If we are discussing *statistical* properties of $\hat{\theta}$—such as deriving whether or not it is unbiased or consistent—then we are necessarily viewing $\hat{\theta}$ as an estimator. On the other hand, if we write something like $\hat{\theta} = 1.73$, then we are clearly denoting a point estimate from a given sample of data. The confusion that can arise by using $\hat{\theta}$ to denote both should be minimal once you have a good understanding of probability and statistics.

SUMMARY

We have discussed topics from mathematical statistics that are heavily relied upon in econometric analysis. The notion of an estimator, which is simply a rule for combining data to estimate a population parameter, is fundamental. We have covered various properties of estimators. The most important small sample properties are unbiasedness and efficiency, the latter of which depends on comparing variances when estimators are unbiased. Large sample properties concern the sequence of estimators obtained as the sample size grows, and they are also depended upon in econometrics. Any useful estimator is consistent. The central limit theorem implies that, in large samples, the sampling distribution of most estimators is approximately normal.

The sampling distribution of an estimator can be used to construct confidence intervals. We saw this for estimating the mean from a normal distribution and for computing approximate confidence intervals in nonnormal cases. Classical hypothesis testing, which requires specifying a null hypothesis, an alternative hypothesis, and a significance level, is carried out by comparing a test statistic to a critical value. Alternatively, a *p*-value can be computed that allows us to carry out a test at any significance level.

KEY TERMS

Alternative Hypothesis
Asymptotic Normality
Bias
Biased Estimator
Central Limit Theorem (CLT)
Confidence Interval
Consistent Estimator
Consistent Test
Critical Value
Estimate
Estimator
Hypothesis Test
Inconsistent
Interval Estimator
Law of Large Numbers (LLN)
Least Squares Estimator
Maximum Likelihood Estimator
Mean Squared Error (MSE)
Method of Moments
Minimum Variance Unbiased Estimator
Null Hypothesis
One-Sided Alternative
One-Tailed Test
Population
Power of a Test
Practical Significance
Probability Limit
p-Value
Random Sample
Rejection Region
Sample Average
Sample Correlation Coefficient
Sample Covariance
Sample Standard Deviation
Sample Variance
Sampling Distribution
Sampling Standard Deviation
Sampling Variance
Significance Level
Standard Error
Statistical Significance
t Statistic
Test Statistic
Two-Sided Alternative
Two-Tailed Test
Type I Error
Type II Error
Unbiased Estimator